Pest Management

Southeast: Insect pests can be major yield-limiting factors in the production of cotton in the U.S. In the southeastern U.S., major insect pests of cotton include thrips (primarily tobacco thrips, *Frankliniella fusca*), bollworm (*Helicoverpa zea*), and stink bugs (multiple species). Recently, however, plant bugs (primarily tarnished plant bug, *Lygus lineolaris*) have increased in importance, and cotton/melon aphid (*Aphis gossypii*) has been implicated in vectoring a new and potentially costly viral pathogen to the crop. An entomology working group involved with entomological research and Extension programming for cotton in the region continued to work collaboratively to address these issues by completing the following objectives for the 2019 regional study.

1. Surveyed for incidence of fields meeting or exceeding threshold for the tarnished plant bug, *Lygus lineolaris*, and any other mirids of economic importance in cotton. A secondary objective was to associate landscape factors with populations. Surveys included two major visits to each field – one pre-bloom and one post-bloom visit.
2. Surveyed for Cotton leafroll dwarf virus (CLRDV- Blue disease) vectored by cotton aphid, *Aphis gossypii*. The extent of the distribution of CLRDV across the Southeast was uncertain. The goal of this objective was to assist in mapping the current distribution of CLRDV across the Southeast by sampling cotton fields from counties not surveyed in 2018.
3. Insecticide efficacy trials for cotton aphid, *Aphis gossypii,* were conducted. As trial opportunities became available, insecticide efficacy trials for cotton aphid were conducted and provided data on potentially slowing the secondary spread of CLRDV with effective materials. Trials included all labeled and unlabeled products that provide control of cotton aphid.
4. Oversprays on 2- and 3-gene Bt cotton for control of bollworm and stink bugs were conducted. Replicated research plots used WideStrike technology, as it is the weakest Bt technology for control of bollworm. Varieties of 2- and 3-gene cotton from Phytogen (similar maturities) were used. Treatments were tested in two separate trials (2-gene and 3-gene Bt cotton) and applied when the greatest impact was expected, simulating what a producer or consultant would use. This research provided valuable information on how to control Bt resistant bollworm while also providing control of stink bugs.

Additionally, tarnished plant bug (TPB) has become a more frequent problem in Southeastern cotton. In 2016, the value of cotton production in Virginia and North Carolina was just under $158 million. Cotton is both labor intensive and costly to manage compared to other field crops produced in these states. Many cotton growers face the decision to continue growing cotton, despite additional expenses associated with managing insect pests, or turn to other crops that require fewer inputs. If TPB infestations continue to increase, similar to the TPB epidemic observed in the Mid-South, the cotton industries in Virginia and North Carolina may be in jeopardy. In an effort to find both effective and sustainable integrated pest management solutions for managing TPB with reduced costs to growers and the environment, a study was replicated in Plymouth, NC, and the Tidewater Agricultural Research and Extension Center in Suffolk, VA, in 2017, 2018, and 2019. The current outcomes of this research included: (1) Determining the effects of an IPM approach on pest density, plant injury (e.g., square retention and internal boll injury), lint yield, and economic return; (2) Evaluating thresholds and sampling techniques; (3) Making beat cloths, provide them to growers, and demonstrate how to scout for insect pests using them.

Silverleaf whitefly is one of the world's most serious insect pests. Unfortunately, whitefly populations have radically increased throughout the southern half of Georgia over the past two years and are directly responsible for widespread economic losses in vegetable and cotton production systems. Entomologists hypothesize that these populations may result from increasingly diverse cropping systems that provide excellent year-round cultivated hosts, increased survival of non-agronomic hosts due to herbicide resistance and climate change, and a lack of suitable natural enemies. Additionally, current crop specific Extension recommendations are likely increasing insecticide resistance selection pressure due to repeated applications of the same active ingredients across crops. Due to direct feeding and indirect factors, such as the transmission of new viruses, economically sustainable production will become untenable if current trends continue and insecticide resistance appears.

To address this emerging issue, funding was provided to support a Postdoctoral Researcher to coordinate whitefly research and Extension efforts across South Georgia cropping systems. The post doc conducted basic research on the host range and annual distribution of whiteflies. These data will catalyze our ability to formulate science-based recommendations and will serve as the basis for development of landscape-wide management plans for whitefly population mitigation going forward.

Mid-South: In 2019, entomology research efforts in the Mid-South focused on bollworms, thrips, and TPB management. Growers are currently able to control these pests but at a high cost. A regional study with locations in Arkansas, Louisiana,